

ExpressionMed

Meghan Sharkus sat at a desk in her University of St. Thomas dorm room in Minnesota surrounded by books and colorful adhesive patches used to secure wearable diabetes-related medical devices. What had started as a high school DECA¹ project, inspired by a childhood



Figure 1. ExpressionMed tape worn on a CGM

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friend diagnosed with diabetes, had grown into a promising business, and orders for her specialized tapes were piling up.

Sharkus had spent much of her senior year of high school developing a new type of medical adhesive that could keep insulin pumps and continuous glucose monitors (CGMs)² secured to skin for up to a month. She used her creativity to make the adhesives fun and stylish, rather than something to be endured (Figure 1). Now, with sales climbing, Sharkus faced some very important decisions—decisions that would determine ExpressionMed's future. One key consideration was: What was the best way to get her tapes into customers' hands?

From Southern Wisconsin to the Medical Device World

Sharkus grew up in Oregon, Wisconsin, a town of around 11,000 people. She had joined DECA as a high school junior, completing a community service project that sparked her interest in business. The following summer, a friend with diabetes complained to Sharkus about the tan lines left by her insulin infusion set, which had to be moved every three days.³ Sharkus thought to herself, "I'll just make her a little craft, and I could sell it on Etsy."

What started as a simple idea—heart-shaped infusion-set tape that would give her friend creative tan lines—turned into serious research and development. Not only were aesthetics an issue, but Sharkus learned that infusion sets were often banged, causing them to come loose at school. She bought every kind of tape available from Walgreens and Alibaba⁴ and realized none of them had

¹ DECA (formerly Distributive Education Clubs of America) is a business-oriented student organization that provides curriculum and hosts competitions.

² Insulin pumps and continuous glucose monitors (CGMs) are medical devices designed for individuals with diabetes. Insulin pumps are wearable devices that continuously deliver insulin to help the body absorb glucose. CGMs measure an individual's blood sugar throughout the day to ensure it is at a safe level.

³ Insulin infusion sets need to be moved to different locations on the body regularly to prevent skin irritation from the tape, avoid infections, and ensure the insulin is being absorbed effectively.

⁴ Alibaba is a Chinese e-commerce business that sells a wide variety of products to consumers and businesses.

the lasting power she needed. She set out to make a new tape; one with a better adhesive and more aesthetically pleasing shapes. “That ended up being what led me to do a year of material science research to have new material that keeps devices on for up to a month,” she said. “That really hadn't been done before.”

During that summer, between junior and senior year, Sharkus attended a six-week entrepreneurship program at the University of Pennsylvania (UPenn) that boosted her confidence, entrepreneurial skills, network, and commitment to creating something tangible.

Because she had no background in materials science, she sought out experts who could help her develop a better adhesive, starting with materials science majors at UPenn and later with materials engineers at 3M, a multinational company based in Maplewood, Minnesota, that produces thousands of products including specialty adhesives (and Post-It notes). “While doing research as a high schooler, I literally got in my car, drove to 3M's office [in neighboring Minnesota], and had a meeting with their innovation team,” she recalled. “It was so unusual for them to have a 17-year-old asking for a meeting that they took it.”

Her first year was devoted entirely to research, business plan writing, and a second DECA competition, with no product sales. Eventually, Sharkus developed a material with promise. Next up would be dealing with manufacturers and building her supply chain. She didn’t just want tape; she wanted tape with unique and creative designs. When Sharkus talked to companies called converters that print on materials like tape, they told her they couldn’t print on her adhesive, but she wouldn’t take no for an answer. Sharkus told one converter, “I just did an entire research sequence, and I know you can.” She added, “‘I will pay you all of my money,’ which was \$1,000 at the time, ‘to run this. If it works, I get that in product. And if it doesn't work, I'm the one that paid for it.’ And it worked.”

By the end of her senior year in high school, Sharkus had won several business competitions, earning six months of mentorship and \$10,000 in funding. The next fall, during her first year of college, she won \$110,000 at the Minnesota Cup business plan competition, which provided the capital she needed to scale her business without investors or debt.

The Big Decision

By 2018, ExpressionMed had established itself as an online retailer of colorful, expressive adhesives for diabetes devices. Sales were growing, and Sharkus was beginning to build a marketing team. However, a critical strategic decision loomed: Should ExpressionMed remain an online-only business, or should she pursue partnerships with brick-and-mortar⁵ pharmacies that carried other diabetes-related materials? Meghan’s research led her to identify the advantages and disadvantages for each approach.

⁵ “Brick and mortar” is a term used to refer to physical retail shops as opposed to online sellers.

The brick-and-mortar route offered potential advantages:

- Getting the adhesive on a pharmacy shelf would offer greater brand visibility and access to customers who do their shopping in person.
- ExpressionMed would have the potential to reach diabetes patients who shopped online and in stores but may not think to search for the product online.
- The tapes would gain credibility just by being on pharmacy shelves.

However, there were drawbacks:

- Retailers typically take a 40% margin,⁶ significantly reducing Sharkus's per-unit profit.
- Different packaging would be required for shelf display versus mail delivery.
- For Class I medical devices, such as the adhesives, selling through major pharmacies would require \$5 million in product liability insurance; the insurance premiums would be an added expense.
- There would be a risk that retailers would create their own store-brand versions of ExpressionMed's adhesives, as they often did with other medical supplies.

The online-only approach also had compelling advantages:

- ExpressionMed could retain its direct relationship with customers.
- It would be easier to quickly adjust marketing, pricing, and product offerings.
- There would be lower overhead costs and less inventory-management complexity without an additional marketing channel.

Sharkus and her team reviewed their data on customer demographics and behavior. They found that contrary to what they had assumed, 75% of their customers were adults buying for themselves, not parents buying for children. Their largest demographic was women between 25 and 45. And this age group is already comfortable with online shopping.

The company had also started selling on Amazon, and its sales data allowed Sharkus to learn more about her potential market. "We asked on Amazon, are we taking customers away from our website where people are paying full price, not at a 15% margin?" Sharkus said. "And they weren't even the same people. The people who were looking for products on Google or from our marketing emails were not the people who were just looking us up on Amazon." The conclusion that ExpressionMed was reaching new customers through Amazon suggested that brick-and-mortar retail might similarly reach an entirely different customer base, potentially justifying the lower margins.

⁶ The retail margin on a good is a percentage of the sales price. The product manufacturer negotiates the sales price with the retail store. In the case of ExpressionMed, the price could be higher than Sharkus charges on her own site, which would help her retain some of her profit margin, or the same, in which case her profit margin would decrease significantly.

Topic 4.3 Case

As Sharkus weighed these factors, she wondered, would the expanded customer reach of brick-and-mortar retail be worth the reduced margins and increased complexity? Or should she focus on optimizing the online experience and expanding digital marketing efforts?

Meghan Sharkus was interviewed in 2025 by a College Board case author for this case study.

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